School of Information Technology and

Advanced PLC programming

Practice

CASCADE

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1. Introduction

In modern process control systems, single-loop control is often not sufficient to handle disturbances and achieve stable operation. For this reason, cascade control is widely used in industrial applications. Cascade control improves system response by using two controllers arranged in a master–slave structure.

This practical work focuses on configuring and testing a Cascade PID Control Module using Honeywell Experion PKS with a C300 controller. The experiment includes PID configuration, history and trend setup, and the implementation of safety logic based on process level conditions.

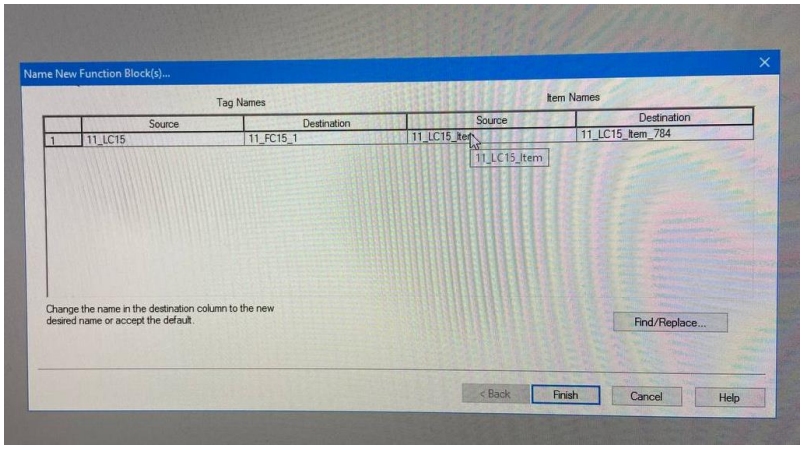
Cascade Control

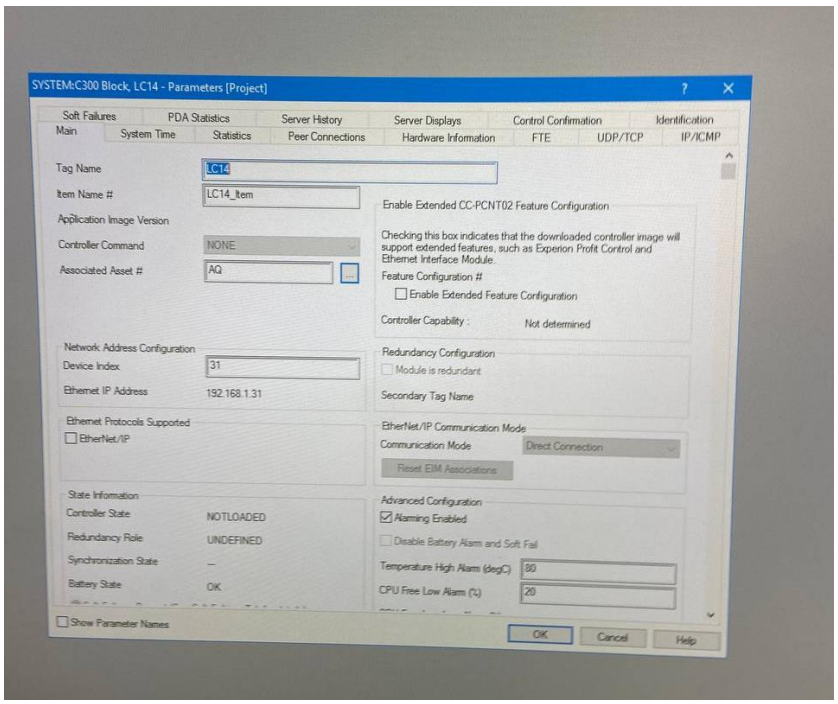
The objective of this configuration is to implement **cascade control** between two controllers.

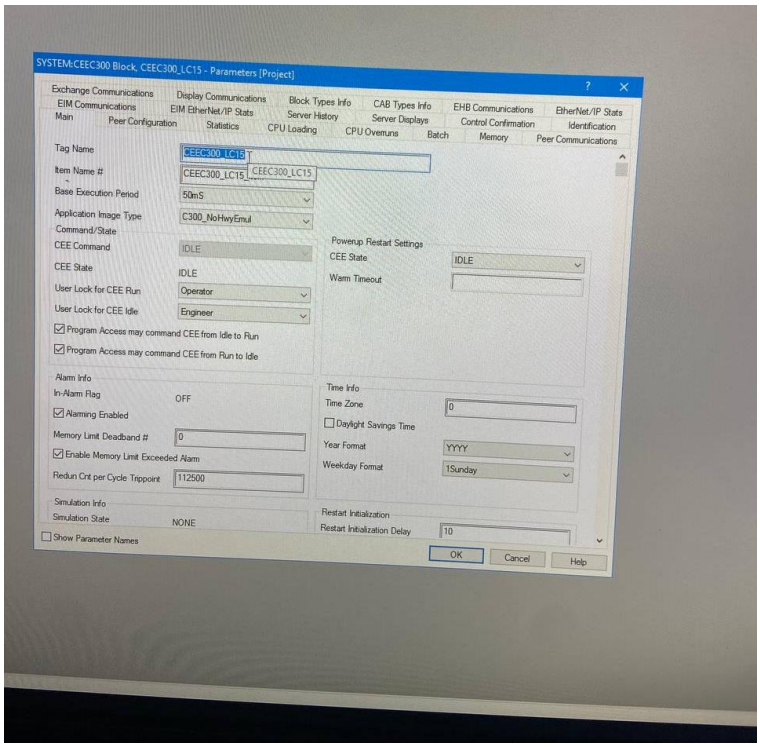
In this setup, the controller **11_FC15** is configured as the **secondary (slave) controller**, while **11_LC15** operates as the **primary (master) controller**.

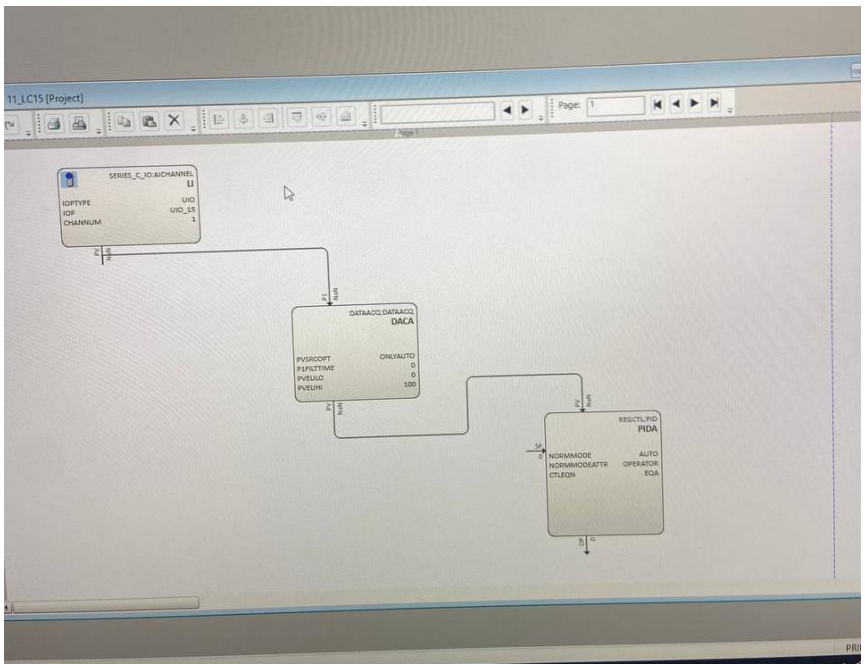
The set point of the secondary controller **11_FC15** is received directly from the output of the primary controller **11_LC15**. For this reason, the **normal mode of the PID block in 11_FC15 is set to CASCADE (CAS)**. This allows the secondary controller to automatically track the set point generated by the primary controller and respond faster to process disturbances.

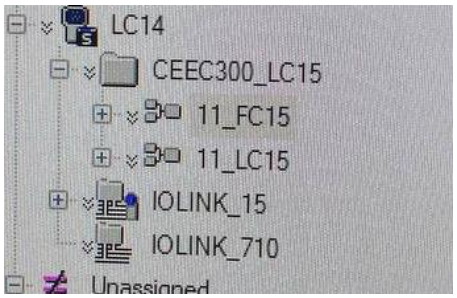
This cascade structure improves overall control performance and process stability.




SYSTEM:C300 Block, LC14 - Parameters [Project]
Soft Failures
PDA Statistics
Server History
Server Displays
Control Confirmation
Identification
Main
System Time
Statistics
Peer Connections
Hardware Information
FTE
UDP/TCP
IP/ICMP
Tag Name
LC14
Item Name #
LC14_Item
Application Image Version
Controller Command
NONE
Associated Asset #
AQ
Enable Extended CC-PCNT02 Feature Configuration
Checking this box indicates that the downloaded controller image will support extended features, such as Experion Profit Control and Ethernet Interface Module.
Feature Configuration #
Enable Extended Feature Configuration
Controller Capability :
Not determined
Network Address Configuration
Device Index
31
Ethernet IP Address
192.168.1.31
Redundancy Configuration
Module is redundant
Secondary Tag Name
Ethernet Protocols Supported
EtherNet/IP
EtherNet/IP Communication Mode
Communication Mode
Direct Connection
Reset EIM Associations
State Information
Controller State
NOTLOADED
Redundancy Role
UNDEFINED
Synchronization State
Battery State
OK
Advanced Configuration
Alarming Enabled
Disable Battery Alarm and Soft Fail
Temperature High Alarm (degC)
80
CPU Free Low Alarm (%)
20
Show Parameter Names
OK
Cancel
Help


SYSTEM:CEEC300 Block, CEEC300_LC15 - Parameters [Project]
Exchange Communications
Display Communications
Block Types Info
CAB Types Info
EHB Communications
EtherNet/IP Stats
EIM Communications
EIM EtherNet/IP Stats
Server History
Server Displays
Control Confirmation
Identification
Main
Peer Configuration
Statistics
CPU Loading
CPU Overruns
Batch
Memory
Peer Communications
Tag Name
CEEC300_LC15
Item Name #
CEEC300_LC15
CEEC300_LC15
Base Execution Period
50mS
Application Image Type
C300_NoHwyEmul
Command/State
CEE Command
IDLE
CEE State
IDLE
User Lock for CEE Run
Operator
User Lock for CEE Idle
Engineer
Program Access may command CEE from Idle to Run
Program Access may command CEE from Run to Idle
Powerup Restart Settings
CEE State
IDLE
Warm Timeout
Alarm Info
In-Alarm Flag
OFF
Alarming Enabled
Memory Limit Deadband #
0
Enable Memory Limit Exceeded Alarm
Redun Cnt per Cycle Trippoint
112500
Time Info
Time Zone
0
Daylight Savings Time
Year Format
YYYY
Weekday Format
1Sunday
Simulation Info
Simulation State
NONE
Restart Initialization
Restart Initialization Delay
10
Show Parameter Names
OK
Cancel
Help





Cascade Control Strategy

In the cascade control structure:

- The **primary controller (11_LC15)** controls the process level
- The **secondary controller (11_FC15)** controls flow
- The output of the primary controller becomes the setpoint of the secondary controller

When the secondary controller is not in CAS mode, the primary controller automatically enters **INIT mode**, indicating that cascade control is temporarily unavailable. This behavior confirms correct cascade configuration.

Identification | Dependencies | Block Pins
Block Preferences | Template Defining
RT Device Status | HART Identification | HART Variables

Channel PV Range

PV Extended High Range	390
PV High Range	385
PV Low Range	0
PV Extended Low Range	-9.6

DATAACQ:DATAACQ Block, DACA - Parameters [Project]

Block Preferences | Template Defining
Main | Alarms | Identification | Dependencies | Block Pins | Configuration Parame

Name : DACA
Execution Order in CM: 30
Description : # T-100 debutanizer bottom
Engr Units : # m3/hr

Process Variable

PV Source Option : (•) ONLYAUTO ( ) ALL
PV Source : AUTO
PV Format : D1
PV Character: NONE

PVEU Range Hi : 385
PVEU Range Lo : 0
PV Extended Hi Limit : 390
PV Extended Lo Limit : -9.6
Low Signal Cut Off: NaN

Clamping/Filtering

Clamping Option : ( ) DISABLE (•) ENABLE
Lag Time : 0 minutes

Main | Algorithm | SetPoint | Output | Alarms

Name: PIDA

Description: # T-100 DEBUTANIZER BO

Engineering Units: # m3/hr

Process Variable

PVEU Range Hi : 385

PVEU Range Low : 0

Manual PV Option : SHEDHOLD

PVRAW Reference : ...

PV Format : D2

☑ Enable Secondary Initialization Option

Execution Order in CM

Mode

Normal Mode

Mode/Mode Attrib

☑ MAN

☑ AUTO

☐ CAS

☐ PROGRAM

☑ Permit Operato

☐ Permit External (Engr Acces

☐ Enable Externa Access / If Per

Execution Order in CM : 30

Mode

Normal Mode CAS

Normal Mode Attribute : NONE

Mode/Mode Attribute Selection

☑ MAN

☑ AUTO

☐ CAS

☐ PROGRAM

Mode : CAS

Mode Attribute : OPERATOR

☑ Permit Operator Mode Changes

☐ Permit External Mode Switching (Engr Access / allows enabling)

☐ Enable External Mode Switching Access / If Permit checked)

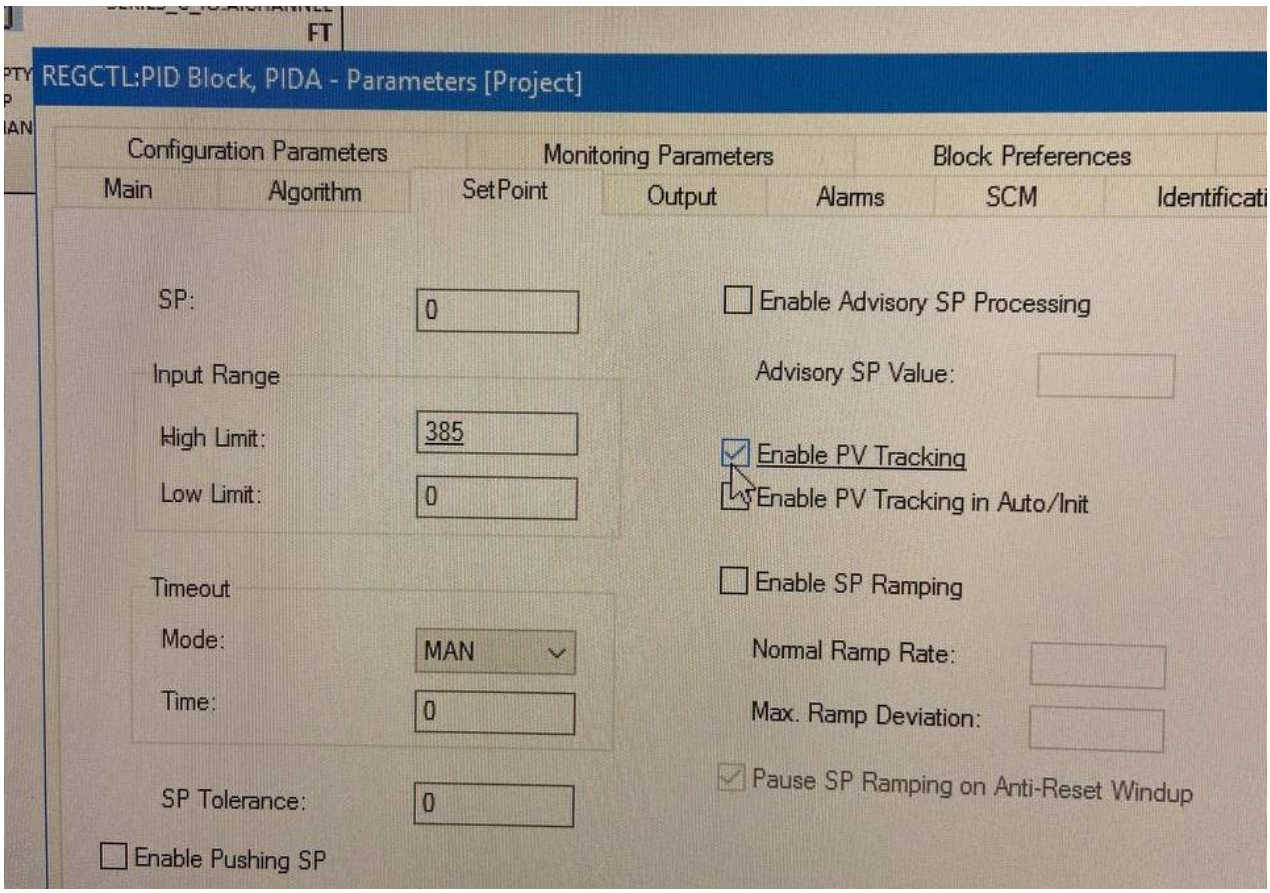
☐ Load and start Mode in Manual

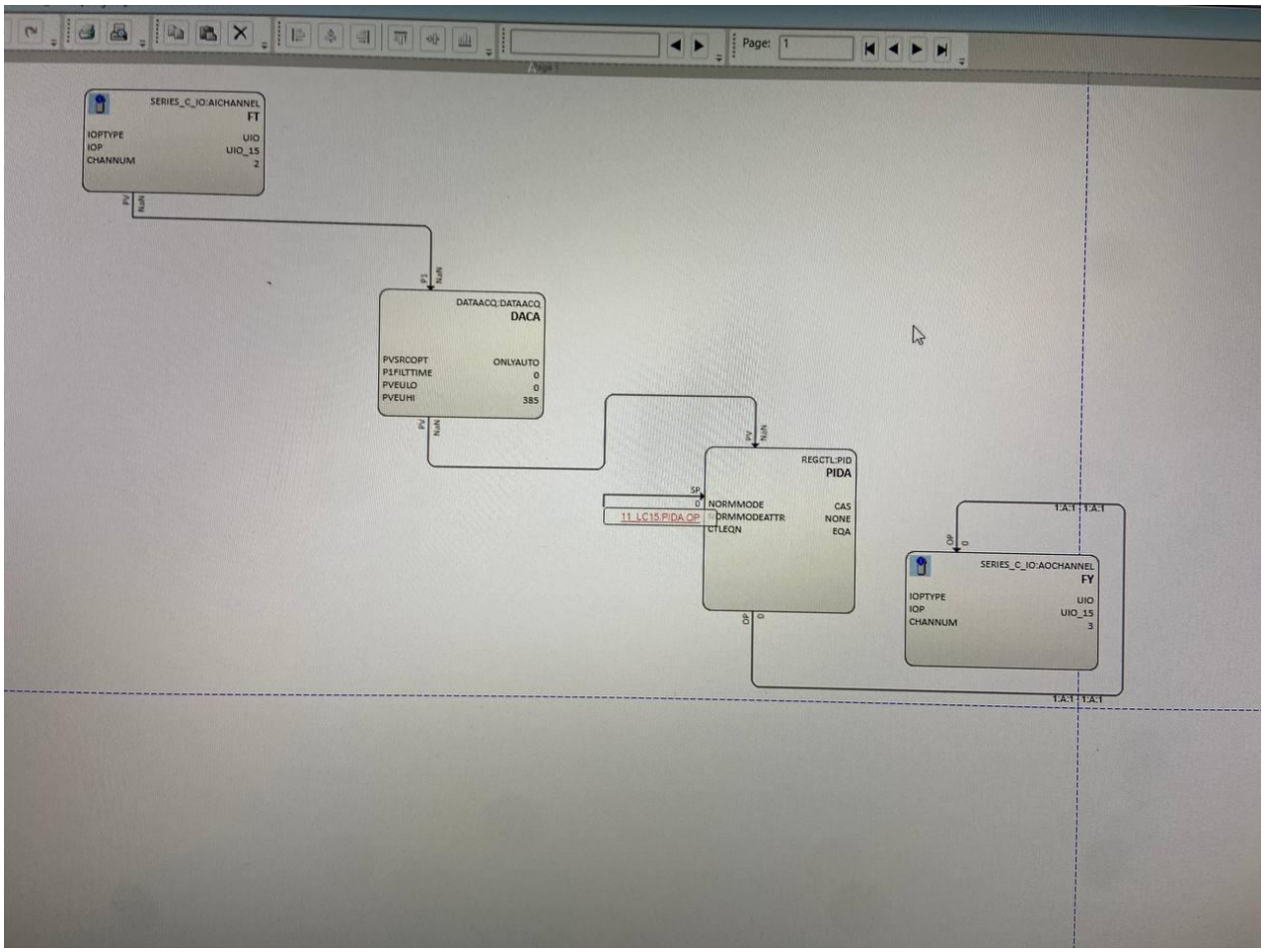
☐ Force Mode Attribute

(Oper

Safety Interlock Option: SHEDHOLD

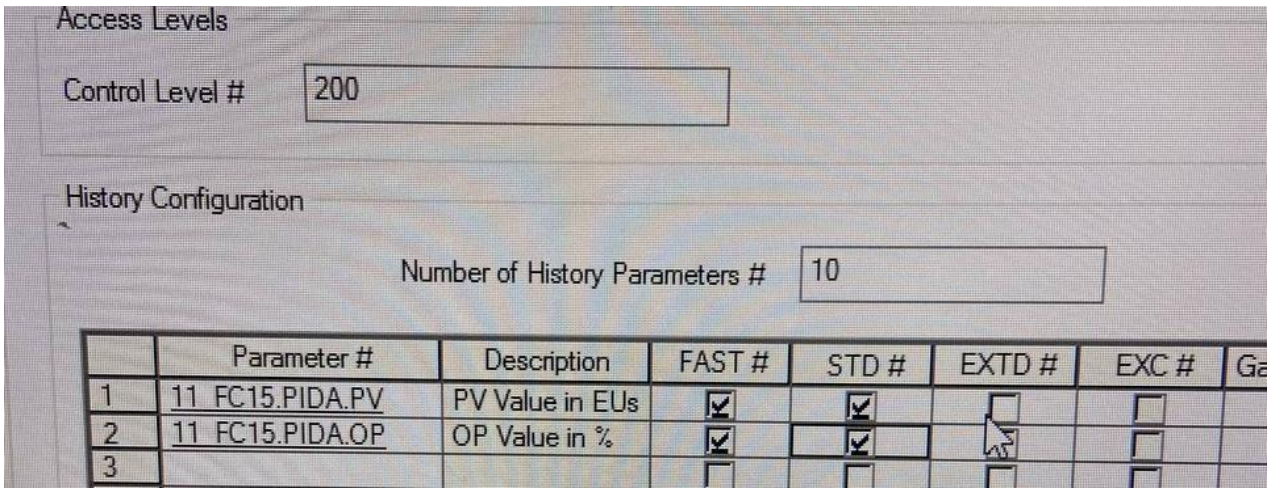
Bad Control Option: NO_SHED





In the **Server Displays** tab:

- **Trend #4, Position #4:** 11_FC15.PIDA.PV
- **Group #11, Position #4:** 11_FC15.PIDA.OP



The following blocks were added:

- **GE (Greater Than or Equal)** block
- **NUMERIC** block

The NUMERIC block stores a fixed comparison value.

- **NUMERIC block**
  - Name: LL_Comparator
  - Actual Value: 8
- **GE block**
  - IN[1]: 11_LC15.DACA.PV
  - IN[2]: LL_Comparator

When the process level is **≥ 8%**, the GE output turns **ON**.

- DACA.PV was connected to GE.IN[1]
- LL_Comparator output was connected to GE.IN[2]

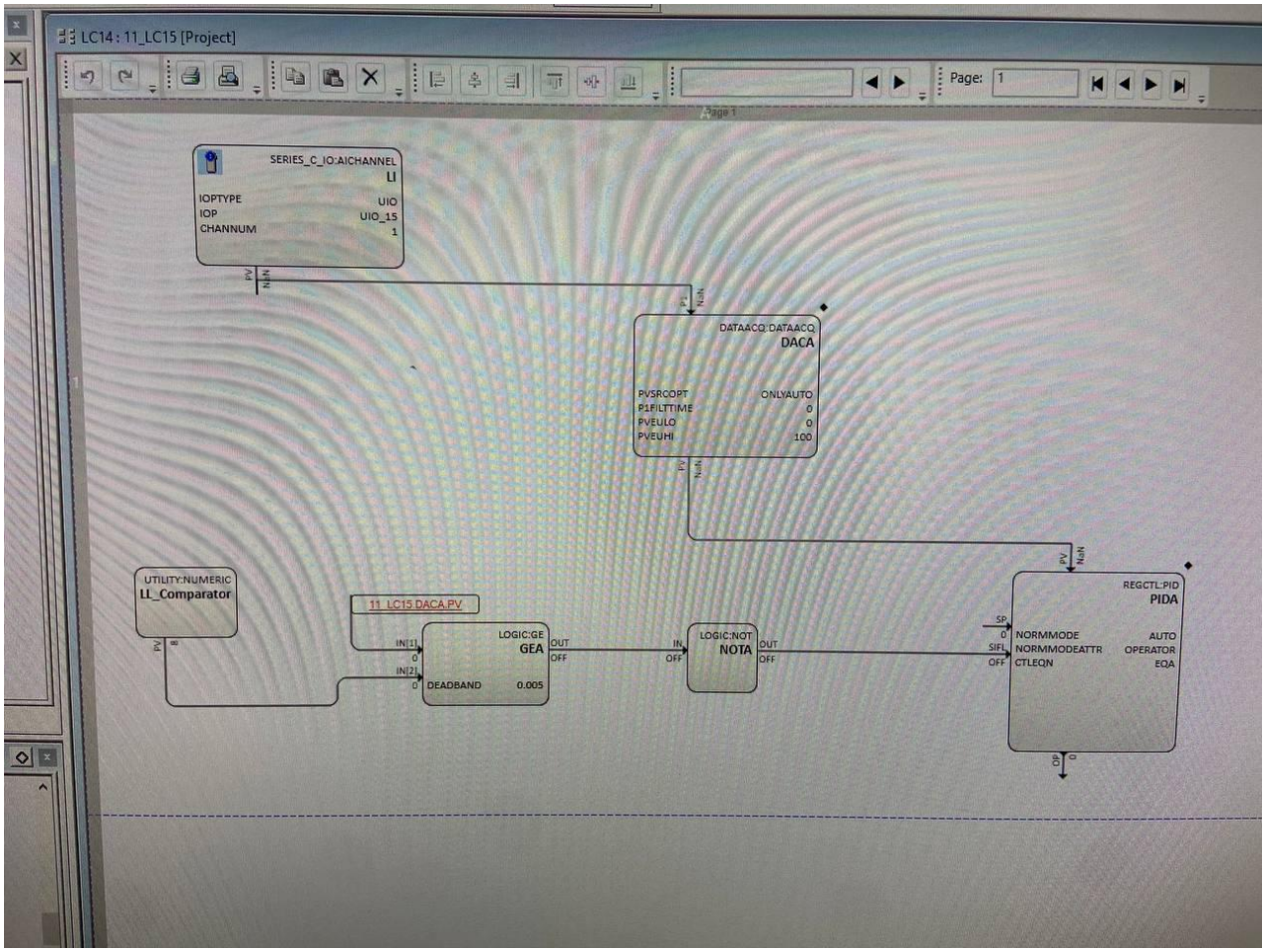
When the level falls below **8%**, the GE output changes state and is used in safety logic.

The **SIFL (Safety Interlock Flag)** pin was added to the PID block of **11_LC15**.

The **Safety Interlock Option (SIOPT)** was set to **SHEDHOLD**.

When SIFL is active:

- Controller switches to **MAN mode**
- Mode attribute is set to **Operator**
- External mode switching is disabled
- Output is held at the last valid value



In this practical work, a complete **cascade PID control system** was successfully configured using Honeywell Experion PKS. The interaction between the primary and secondary controllers functioned correctly, and safety logic based on process level was properly implemented.

The experiment demonstrated the advantages of cascade control, including improved stability, faster response to disturbances, and enhanced process safety. This lab provided valuable hands-on experience in advanced PID control strategies used in real industrial systems.